I introduce my design of the Sovella table height adjustment mechanism. Please see all technical calculations attached. As per Sovella's requirement, my mechanism works without external energy supply (electricity, pressure air, hydraulic etc.).

My design is based on the principle of Ball Screw drive. Ballscrews are never self-locking. A holding brake is necessary to sustain loads as Ballscrews can easily convert thrust to torque and backdrive a gearbox or motor or other drive system elements. The effectiveness of this system is 95%. Low coefficient of friction (0.01) maximizes the use of the system. Low coefficient of friction requires less force on the handle to lift 500kg. The design requires less care.

In this design, I tried to use as many standard components as I could. Ball Screw drive could be manufactured or purchased as standard SKF series SD/BD. Bevel gear is designed so that you can use inexpensive materials. For bevel gear production material with the yield strength of 300 Mpa is enough.

The construction of the table is easy to modify to match the required size. It is enough to change the "Connection shaft" connecting the two Ball screw drive mechanisms and the countertop. And it is also easy to replace the handle on the electric motor.

Specifications:
The maximum load on the Ball Screw drive - 2 x 250 kg (500 kg);
Maximum speed of handle rotation - 90rpm;
The maximum linear lift speed - 15mm/s;
Ballscrew lead – 10мм;
Height adjustment - from 650mm to 1050mm.